

Newman-Penrose Formalism Calculations

Stephani et al., Equation (15.12_k1)

Metric

$$ds^2 = -P(r) dt \otimes dt + \frac{1}{P(r)} dr \otimes dr + r^2 d\theta \otimes d\theta + r^2 \sin^2(\theta) d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \theta, \phi)$$

Metric Functions

$$P = 1 - \frac{1}{3} \Lambda r^2 + \frac{e^2}{r^2} - \frac{2m}{r} + 1$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \sqrt{P(r)} \right) dt + \left(\frac{\sqrt{2}}{2 \sqrt{P(r)}} \right) dr$$

$$n = \left(\frac{1}{2} \sqrt{2} \sqrt{P(r)} \right) dt + \left(-\frac{\sqrt{2}}{2 \sqrt{P(r)}} \right) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r \right) d\theta + \left(\frac{1}{2} i \sqrt{2} r \sin(\theta) \right) d\phi$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r \right) d\theta + \left(-\frac{1}{2} i \sqrt{2} r \sin(\theta) \right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\sqrt{P(r)}} \right) dt + \left(\frac{1}{2} \sqrt{2}\sqrt{P(r)} \right) dr$$

$$n = \left(-\frac{\sqrt{2}}{2\sqrt{P(r)}} \right) dt + \left(-\frac{1}{2} \sqrt{2}\sqrt{P(r)} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2r} \right) d\theta + \left(\frac{i\sqrt{2}}{2r\sin(\theta)} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}}{2r} \right) d\theta + \left(-\frac{i\sqrt{2}}{2r\sin(\theta)} \right) d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\sqrt{P(r)}} \partial_t + \frac{1}{2} \sqrt{2}\sqrt{P(r)} \partial_r$$

$$\Delta = -\frac{\sqrt{2}}{2\sqrt{P(r)}} \partial_t - \frac{1}{2} \sqrt{2}\sqrt{P(r)} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2r} \partial_\theta + \frac{i\sqrt{2}}{2r\sin(\theta)} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}}{2r} \partial_\theta - \frac{i\sqrt{2}}{2r\sin(\theta)} \partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}\sqrt{P\left(r\right)}}{2\,r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\sqrt{P\left(r\right)}}{2\,r}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = -\frac{\sqrt{2}\cos\left(\theta\right)}{4\,r\sin\left(\theta\right)}$$

$$\beta = \frac{\sqrt{2}\cos\left(\theta\right)}{4\,r\sin\left(\theta\right)}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}\frac{\partial}{\partial r}P\left(r\right)}{8\sqrt{P\left(r\right)}}$$

$$\gamma = \frac{\sqrt{2}\frac{\partial}{\partial r}P\left(r\right)}{8\sqrt{P\left(r\right)}}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = -\frac{P(r)}{4r^2} + \frac{1}{4r^2} + \frac{1}{8} \frac{\partial^2}{(\partial r)^2} P(r)$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = -\frac{\frac{\partial}{\partial r} P(r)}{6r} - \frac{P(r)}{12r^2} + \frac{1}{12r^2} - \frac{1}{24} \frac{\partial^2}{(\partial r)^2} P(r)$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{\frac{\partial}{\partial r}P(r)}{6\,r} + \frac{P(r)}{6\,r^2} - \frac{1}{6\,r^2} + \frac{1}{12}\frac{\partial^2}{(\partial r)^2}P(r)$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"